Google Cloud Data Analyst Certification

Yesh Gowda
4VV22CS190

Introduction to Google Cloud Data Analytics

The Google Cloud Data Analytics course is designed to introduce learners to core tools and techniques for processing, analyzing, and visualizing data on GCP. It covers key services like BigQuery for scalable data querying, Cloud Storage for secure data storage, and Cloud Dataflow for real-time processing.

Learners also explore data visualization with Looker and Data Studio, and integrate AI/ML using Vertex AI. The course provides practical skills to build data pipelines, perform analytics at scale, and make data-driven decisions using Google Cloud.

Objectives of the Course:

Equip Learners with Knowledge

Gain the theoretical understanding and practical skills needed to design, build, and manage data analytics solutions on Google Cloud Platform.

Hands-on Skills Development

Learn to collect, store, process, analyze, and visualize data using GCP tools such as BigQuery, Cloud Storage, Dataflow, and Looker.

Enable Data-Driven Decisions

Develop the ability to leverage data for informed decision-making, optimize business performance, and prepare for real-world data analytics roles.

Prepare for a Cloud Data Analyst Job

Review Key Skills

Review proficiency in BigQuery, data pipeline building, and data visualization, aligning with cloud data analyst expectations.

Capstone Project

Apply learned skills to a complete data analytics scenario, creating a portfolio-ready project for real business problems.

Job Preparation

Gain resume tips, interview guidance, and resources to explore career opportunities in cloud data analytics.

Course Modules: Introduction to Data Analytics

Cloud Computing Fundamentals

Introduces IaaS, PaaS, SaaS, benefits of cloud for data analytics, and foundational GCP services.

Cloud-Based Analytics

Explores scalability, speed, and flexibility of cloud analytics, covering GCP tools like BigQuery, Dataflow, Dataprep, and Looker.

The Data Lifecycle

Explains data's journey from ingestion to visualization, aligning GCP services with each stage for real-world workflows.

Role of a Cloud Data Analyst

Focuses on responsibilities and skills, emphasizing BigQuery for querying and transforming raw data into meaningful insights.

Key Components :

- **BigQuery : A serverless data warehouse that enables fast and efficient data analysis.**
- **Cloud Dataproc: A managed service for running Apache Spark and Hadoop clusters.**
- **Cloud Dataflow: A serverless service for batch and streaming data processing.**

Google Dataflow

- **Cloud Dataprep: A tool that streamlines data preparation and cleansing for analysis**

Lab Activity: Navigate BigQuery

This lab provides hands-on experience with the BigQuery interface, focusing on exploring and querying datasets. Learners will navigate the GCP Console, open BigQuery, and understand dataset structure and schema.

Tasks include previewing and exploring tables in the look_gcda dataset, and locating and accessing the NCAA Basketball Public Dataset to run basic SQL queries for sports statistics and trend analysis.

Data Management and Storage :

Cloud Storage Introduction

Covers concepts of storing and managing data in the cloud, highlighting advantages like scalability, reliability, and cost-efficiency with Google Cloud.

Data Organization

Explores data structuring using datasets, tables, rows, columns, and schemas, emphasizing efficient querying and analytics in BigQuery.

Finding Data

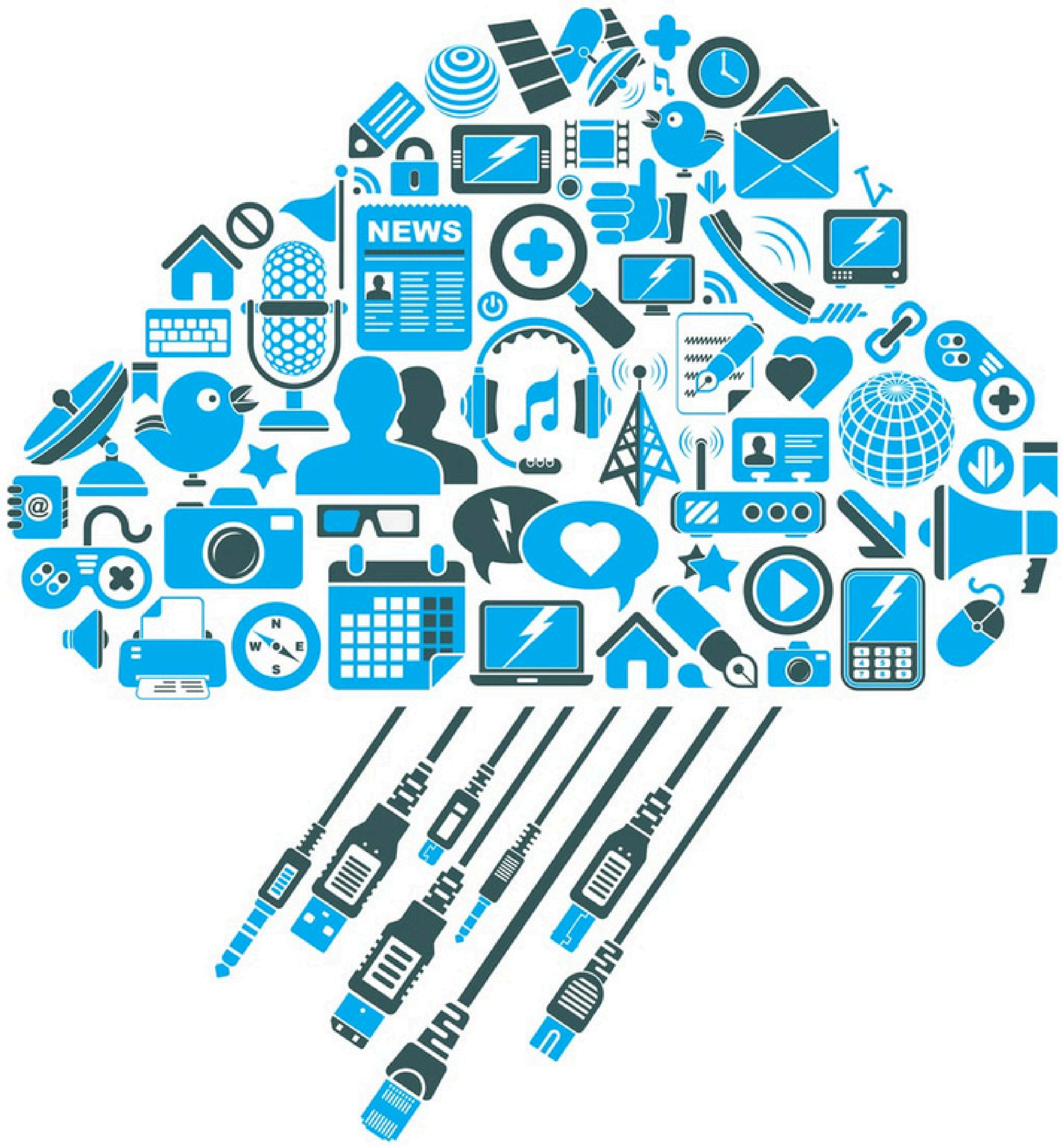
Practical steps for locating datasets in Google Cloud, navigating the GCP Console and BigQuery interface to explore public and private data sources.

Accessing Data

Studies methods to access and query data, including SQL in BigQuery, filtering fields, and optimizing access to large-scale data efficiently.

Data Transformation in the Cloud:

This module introduces data transformation and its importance in analytics workflows, explaining how cloud tools simplify cleaning, enriching, and restructuring data. Learners handle raw data by building automated data pipelines using tools like Cloud Dataflow and SQL-based workflows for efficient real-time or batch processing.



The course also covers optimization strategies for cloud data transformation, including minimizing cost, improving query performance, and structuring data effectively for downstream analysis.

Analyzing and Visualizing Data in Looker:

- Looker: A business intelligence (BI) tool that connects directly to your database for real-time data exploration and analysis.
- Building and sharing interactive dashboards.
- Using Looker's data exploration tools: filters, pivots, and visualizations.

Prepare Data for ML APIs on Google Cloud:

- Data preprocessing: Cleaning, transforming, and normalizing data.
- Automate data preparation for analysis and machine learning.
- Integrating data with Google Cloud’s AI and ML APIs.
- Data quality considerations: Handling missing data, outliers, and feature scaling.

Prepare Data for Looker Dashboards and Reports:

- ETL (Extract, Transform, Load) processes: Best practices and tools.
- Structuring data for ease of use in Looker: Creating clean, usable datasets.
- Data optimization: Indexing, partitioning, and query optimization

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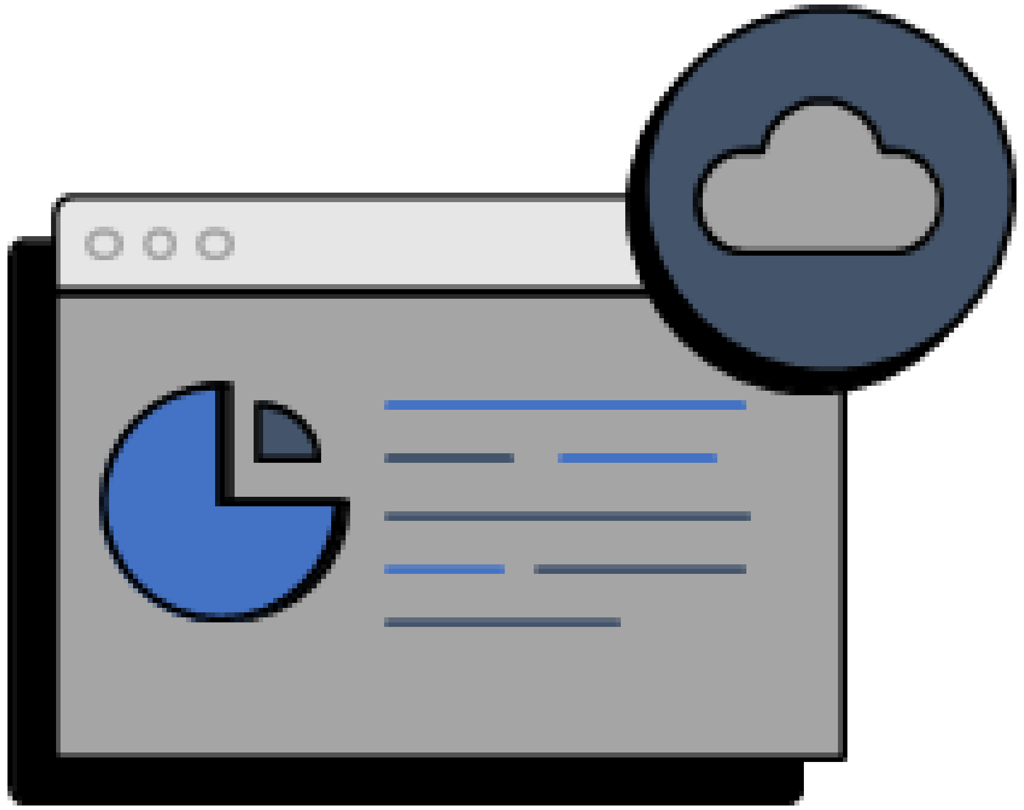
- Filtering: Narrow down data by applying conditions based on dimensions.
- Sorting & Pivoting: Organize data for better insights, with options to pivot tables for cross-analysis.



Developing Data Models with LookML:

- LookML: Modeling language in Looker to define dimensions, measures, and business logic.

- Enables consistent, reusable data models.

- Creating and managing models: Exploring and creating views, explores, and joins.

- Best practices for data modeling: Ensuring accuracy, performance, and scalability.

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Conclusion:

- This learning path offered a comprehensive foundation for aspiring Data Analysts, combining theoretical knowledge with practical, real-world experience.
- Through a series of carefully selected on-demand courses, interactive labs, and skill badges, you will develop proficiency in leveraging Google Cloud technologies critical to the Data Analyst role.
- Upon completion, we were well-equipped to gather, analyze, and interpret data, uncover trends, and generate actionable insights. This expertise will enable us to address complex business challenges and contribute to informed decision-making within the organization.